

FAST NUCES Islamabad
Department of Computer Science

Probability and Statistics — MT-2005
Online News Popularity
Linear Regression & Classification

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1. Introduction & Motivation

In the digital media landscape, predicting the popularity of an article before or shortly after publication offers significant value to publishers, marketers, and content creators. Understanding which features drive user engagement helps organizations optimize content strategy, improve audience targeting, and maximize reach.

This project uses the UCI Online News Popularity Dataset to examine how article characteristics, including content length, sentiment, topic category, keyword quality, and publishing time influence the number of times an article is shared on social media. We apply three core analytical approaches:

- **Linear Regression:** continuous prediction of share counts (log-transformed)
- **Logistic Regression:** binary classification of articles as High or Low popularity
- **Hypothesis Testing:** statistical inference on 5 selected features

Target Variable:

- **shares:** the number of times an article was shared on social media platforms, used as a proxy for article popularity.

2. Dataset Overview

The UCI Online News Popularity Dataset contains information about online news articles collected from Mashable (a content aggregation platform). Each row represents one published article.

```
2. Data Loading & Initial Inspection

df = pd.read_csv('/content/OnlineNewsPopularity.csv')
df.columns = df.columns.str.strip()
print(f'Shape: {df.shape}')
df.head()
```

	url	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_tokens	num_hrefs	num_self_hrefs	num_imgs	...	min_positive
0	http://mashable.com/2013/01/07/amazon-instant...	731.0	12.0	210.0	0.003594	1.0	0.815385	4.0	2.0	1.0	...	...
1	http://mashable.com/2013/01/07/ap-samsung-spon...	731.0	9.0	255.0	0.004743	1.0	0.791940	3.0	1.0	1.0	...	...
2	http://mashable.com/2013/01/07/apple-40-billio...	731.0	9.0	211.0	0.075130	1.0	0.003806	3.0	1.0	1.0	...	...
3	http://mashable.com/2013/01/07/astonaut-note...	731.0	9.0	531.0	0.503788	1.0	0.005635	9.0	0.0	1.0	...	...
4	http://mashable.com/2013/01/07/att-u-verse-apps/	731.0	13.0	1072.0	0.415640	1.0	0.540890	19.0	19.0	20.0	...	...

```
5 rows x 61 columns
```

```
print('Column dtypes:')
print(df.dtypes.value_counts())
print(f'Missing values: {df.isnull().sum().sum()}')
print(f'Duplicate rows: {df.duplicated().sum()}')

print('Column Names: {df.columns}')

Column dtypes:
float64    59
object      1
int64      1
Name: count, dtype: int64

Missing values: 0
Duplicate rows: 0
Column Names: Index(['url', 'timedelta', 'n_tokens_title', 'n_tokens_content', 'n_unique_tokens', 'n_non_stop_words', 'n_non_stop_unique_tokens', 'num_hrefs', 'num_self_hrefs', 'num_imgs', 'num_videos', 'average_token_length', 'num_keywords', 'data_channel_is_lifestyle', 'data_channel_is_entertainment', 'data_channel_is_bus', 'data_channel_is_sports', 'data_channel_is_tech', 'data_channel_is_world', 'kw_min_min', 'kw_max_min', 'kw_avg_min', 'kw_min_max', 'kw_max_max', 'kw_avg_max', 'kw_min_avg', 'kw_max_avg', 'kw_avg_kw', 'self_reference_min_shares', 'self_reference_max_shares', 'self_reference_avg_shares', 'weekday_is_monday', 'weekday_is_tuesday', 'weekday_is_wednesday', 'weekday_is_thursday', 'weekday_is_friday', 'weekday_is_saturday', 'weekday_is_sunday', 'is_weekend', 'lda_0', 'lda_1', 'lda_2', 'lda_3', 'lda_4', 'global_subjectivity', 'global_sentiment_polarity', 'global_rate_positive_words', 'rate_negative_words', 'rate_positive_words', 'rate_negative_polarity', 'rate_positive_polarity', 'min_positive_polarity', 'max_positive_polarity', 'avg_negative_polarity', 'min_negative_polarity', 'max_negative_polarity', 'title_sentiment_polarity', 'abs_title_subjectivity', 'abs_title_sentiment_polarity', 'shares'],
      dtype: object)
```

2.1 Feature Categories

The 55 input features fall into six broad groups:

- **Content metrics:** word count, unique tokens, links, images, videos
- **Keyword statistics:** min/max/avg shares of best and worst keywords (kw_*)
- **Topic channels:** binary flags for Lifestyle, Entertainment, Business, Social Media, Tech, World
- **Temporal features:** weekday indicators (Mon–Sun) and is_weekend flag
- **LDA topic scores:** 5 LDA topic-closeness scores (LDA_00 to LDA_04)
- **Sentiment features:** subjectivity, polarity (global and title-level)

3. Data Preprocessing

```
3. Data Preprocessing

Invalid Data Removal

• Removing url and timedelta because they are not helpful for prediction.
• Rows with Shares < 0, because in real world, shares can not be negative (no value found)
• Rows with n_unique_tokens > 1 represent data entry errors (the feature is defined as a ratio). Only one row found

df.drop(columns=['url', 'timedelta'], inplace=True)
print(f'After dropping url & timedelta: {df.shape}')

print(f'\nRows with shares <= 0: {(df["shares"] <= 0).sum()}')
df = df[df['shares'] > 0].reset_index(drop=True)

print(f'\nn_unique_tokens > 1: {(df["n_unique_tokens"] > 1).sum()}')
#removing rows having token outside range
df = df[(df['n_unique_tokens'] <= 1) & (df['n_non_stop_unique_tokens'] <= 1)].reset_index(drop=True)
print(f'After filtering token ratios: {df.shape}')

... After dropping url & timedelta: (39644, 59)

Rows with shares <= 0: 0

n_unique_tokens > 1: 1
After filtering token ratios: (39643, 59)

#removing those features which have same value in each row they are repetitive and of no use in prediction
selector = VarianceThreshold(threshold=0.0)
selector.fit(df.drop(columns=['shares']))
zero_var_cols = df.drop(columns=['shares']).columns[~selector.get_support()].tolist()
print(f'Zero-variance features (redundant): {zero_var_cols}')

# finding perfect correlation pairs
corr_matrix = df.drop(columns=['shares']).corr().abs()
upper_tri = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))
high_corr_pairs = [(col, row, upper_tri.loc[row, col])
                  for col in upper_tri.columns
                  for row in upper_tri.index
                  if upper_tri.loc[row, col] > 0.90]
print(f'\nHighly correlated pairs (>0.90):')
for c1, c2, val in high_corr_pairs:
    print(f' {c1} <-> {c2}: {val:.3f}')

... Zero-variance features (redundant): []

Highly correlated pairs (>0.90):
n_non_stop_unique_tokens <-> n_unique_tokens: 0.938
average_token_length <-> n_non_stop_words: 0.944
kw_avg_min <-> kw_max_min: 0.941

#removing redundant highly correlated features
to_drop_corr = ['n_non_stop_unique_tokens', 'n_non_stop_words', 'kw_max_min']
df.drop(columns=to_drop_corr, inplace=True)
print(f'Dropped: {to_drop_corr}')
print(f'Remaining shape: {df.shape}')

Dropped: ['n_non_stop_unique_tokens', 'n_non_stop_words', 'kw_max_min']
Remaining shape: (39643, 56)
```

3.1 Invalid Data Removal

Three cleaning steps were applied before modeling:

- **Dropped url and timedelta:** these are identifiers, not predictive features.
- **Removed rows with shares ≤ 0 :** shares cannot be negative in the real world (0 rows affected).
- **Removed rows where n_unique_tokens > 1:** this ratio feature is defined in [0,1]; values above 1 indicate data entry errors (1 row removed).

3.2 Redundant Feature Removal

Two types of redundant features were identified and removed:

- **Zero-variance features:** columns with identical values in every row contribute nothing to prediction. None were found.
- **Highly correlated features ($|r| > 0.90$):** when two features carry nearly identical information, one is dropped to reduce multicollinearity. The VarianceThreshold and correlation matrix approach was used.

3.3 Missing Values & Duplicates

- No missing values were found in the dataset.
- No duplicate rows were found.

4. Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

Key statistical properties of the target variable shares:

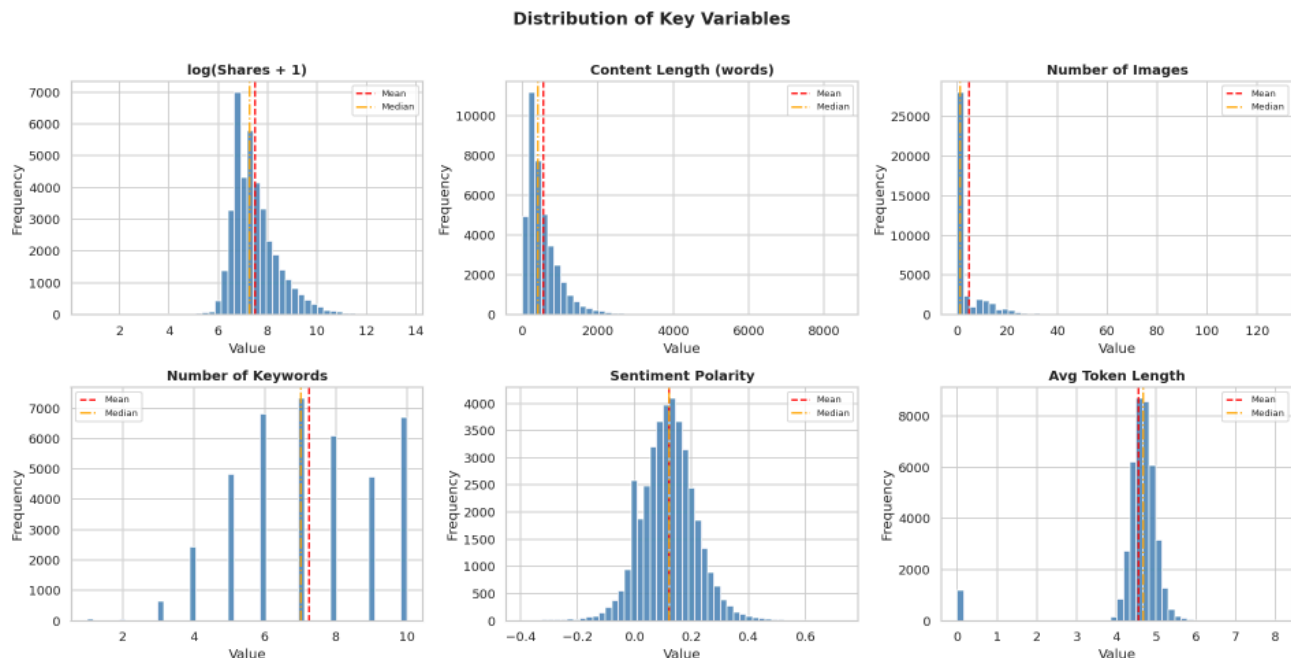
The mean is more than double the median, and the maximum is orders of magnitude larger, classic signs of a heavy right tail. A few viral articles dominate the distribution, which motivates log-transformation of the target for regression.

```
[54] ✓ Os print('Basic statistics of target variable (shares):')
df['shares'].describe()

--- Basic statistics of target variable (shares):
shares
count    39644.000000
mean     3395.380184
std      11626.950749
min       1.000000
25%      946.000000
50%     1400.000000
75%     2800.000000
max     84300.000000
dtype: float64
```

4.2 Distribution of Key Variables

Histograms reveal that shares is heavily right-skewed. When plotted on a log scale, the distribution becomes approximately bell-shaped, confirming that $\log(1 + \text{shares})$ is an appropriate transformation. Other content features (word count, image count) also exhibit right skew.



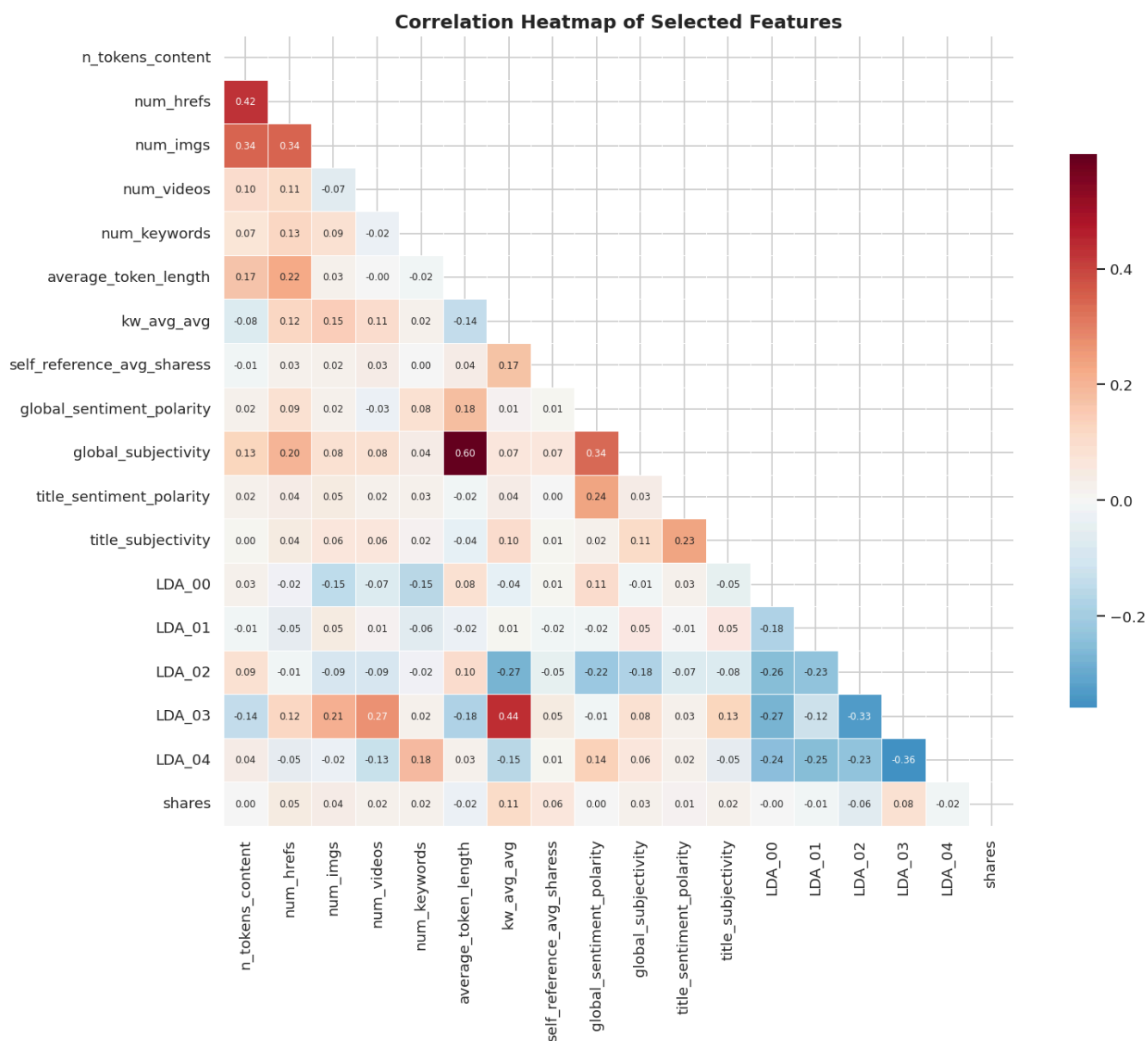
4.3 Correlation Heatmap

A Pearson correlation heatmap was computed for a selected subset of features plus the target.

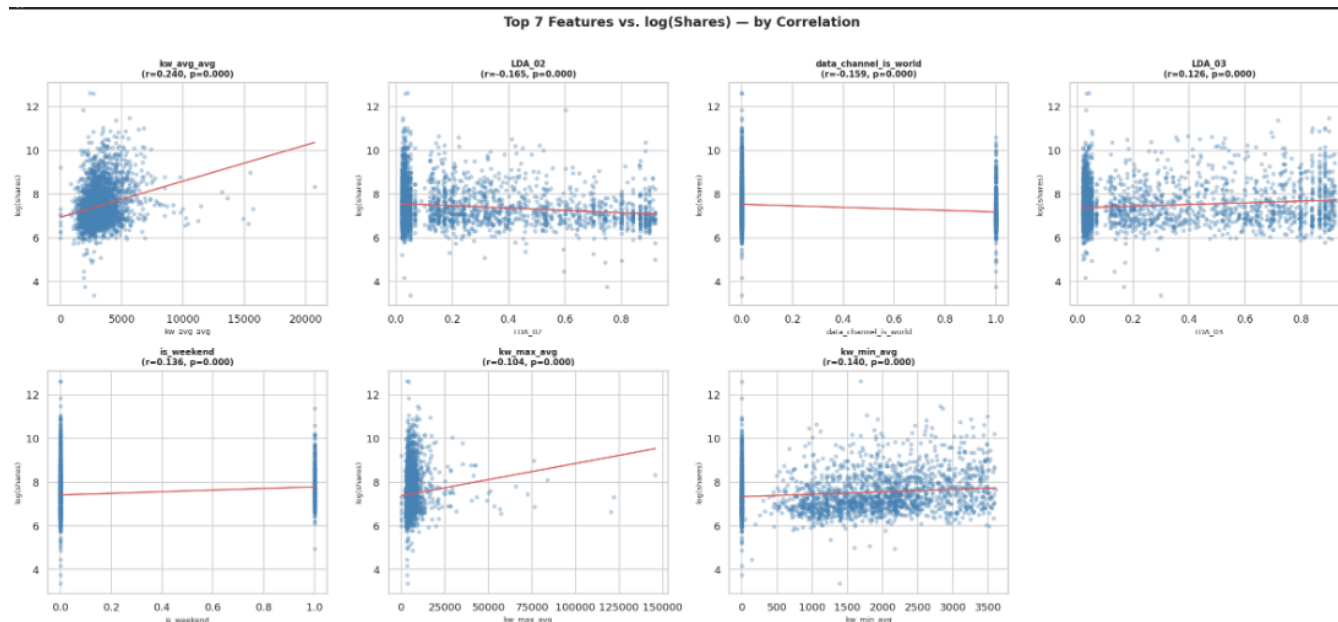
Key observations:

- *average_token_length* and *global_subjectivity*: strongest correlation in the heatmap ($r = 0.60$), articles using longer words tend to be more subjective in writing style.
- *global_subjectivity* and *global_sentiment_polarity*: moderate positive correlation ($r = 0.34$), more subjective articles also tend to be more positive in tone.

- *title_sentiment_polarity* and *global_sentiment_polarity*: moderate correlation ($r = 0.24$) , articles with positive titles tend to have a positive overall tone as well.
- LDA_03 and kw_avg_avg: moderate positive correlation ($r = 0.44$)
- LDA_03 ↔ LDA_04 = -0.36, LDA_02 ↔ LDA_03 = -0.33 — LDA features negatively correlated due to compositional constraint (sum to 1)
- num_hrefs and num_imgs: $r = 0.34$
- *shares* row: max $r = 0.11$, all weak

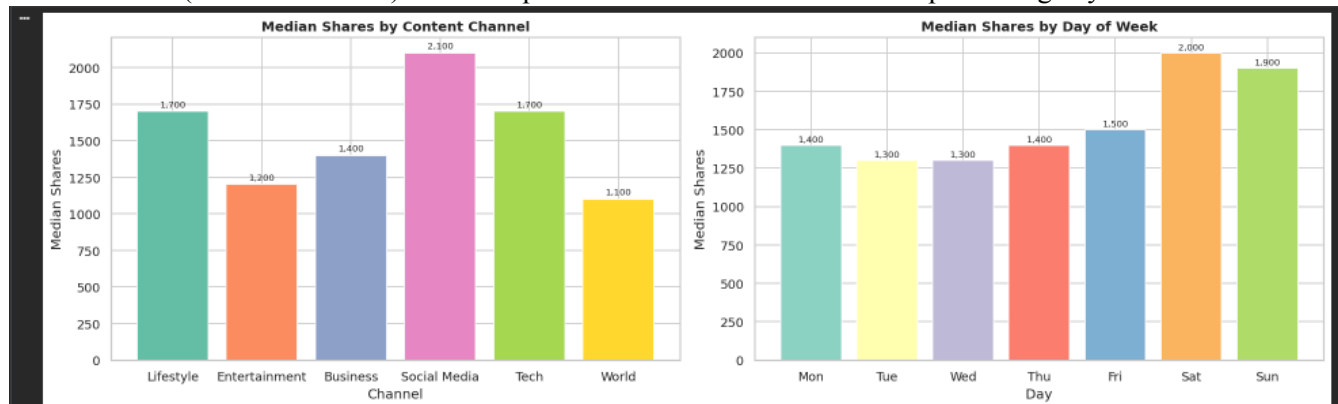


Even the strongest individual correlations with shares are modest (~ 0.1 – 0.2), confirming that no single feature dominates and that linear models will have limited R^2 .



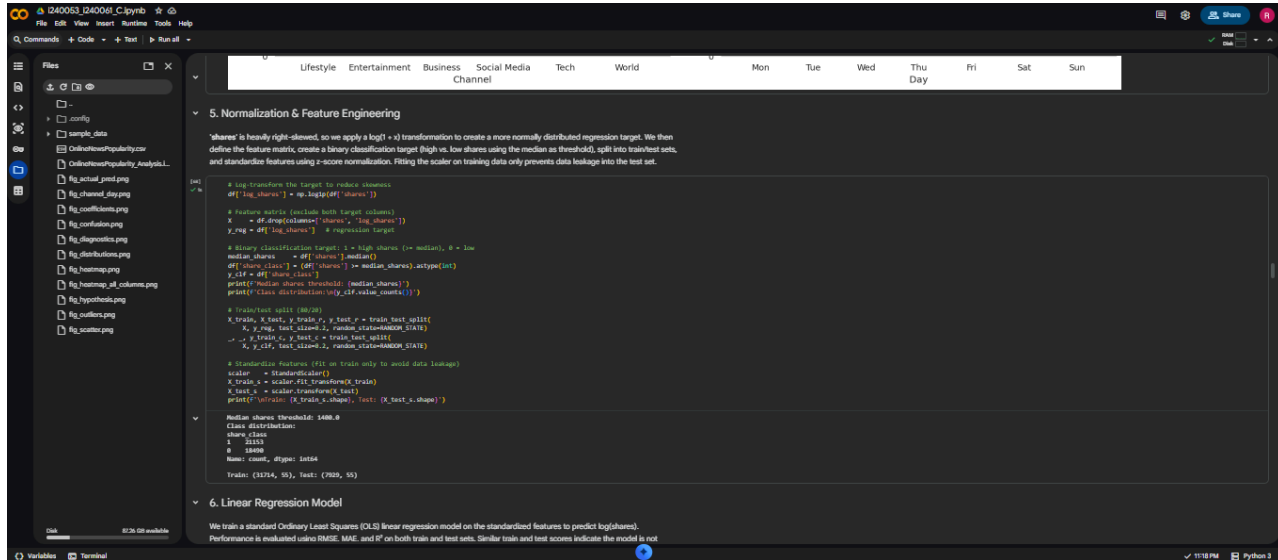
4.4 Channel & Day-of-Week Analysis

Median shares (robust to outliers) were compared across content channels and publishing days:



Weekend articles (Saturday & Sunday) consistently achieve higher median shares than weekday articles, suggesting audience browsing behavior peaks on weekends.

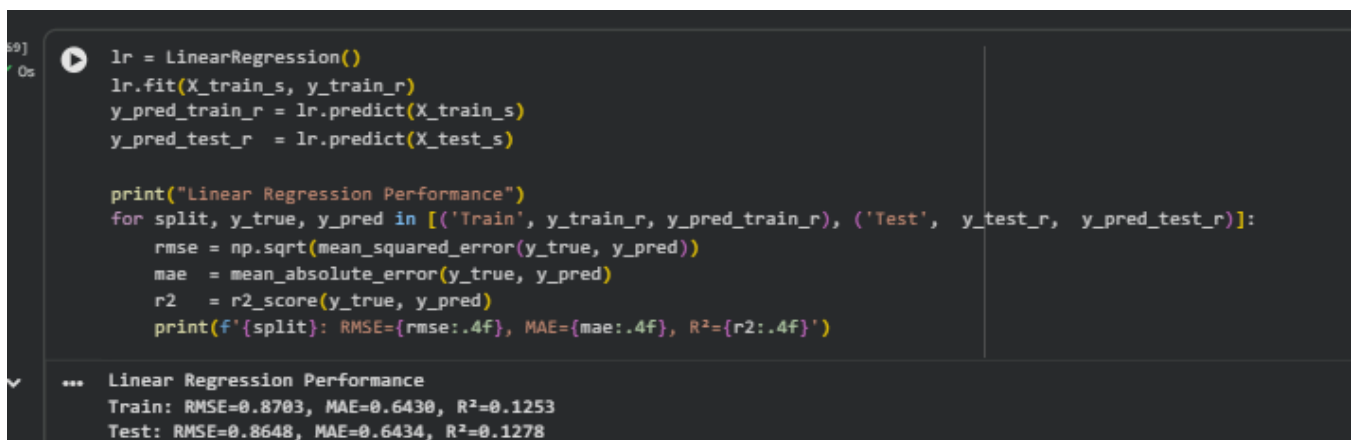
5. Normalization & Feature Engineering



- **Log transformation of target:**
shares $\Rightarrow \log(1 + \text{shares})$ to reduce skewness and stabilize variance for linear regression
- **Binary classification target:**
articles with shares $\geq 1,400$ (median) labelled High (1); below labelled Low (0)
- **Train/test split:**
80% training, 20% test with random_state=42 for reproducibility
- **StandardScaler (z-score normalization):**
fitted on training data only and applied to both train and test sets to prevent data leakage

6. Linear Regression Model

An Ordinary Least Squares (OLS) linear regression model was trained on the 55 standardized features to predict $\log(\text{shares})$. Performance metrics on both train and test sets:



The near-identical train and test scores confirm the model is not overfitting. The R^2 of 0.1278 indicates the model explains about 13% of variance in $\log(\text{shares})$ – modest but consistent with published results on this dataset, as social media virality is driven by many factors not captured in the features (influencer activity, platform algorithms, timing).

6.1 Mean Baseline vs. Fitted Model

A mean baseline model always predicts the training mean for every sample – the simplest possible approach requiring no features. Comparing against it quantifies the real value added by linear regression:

The linear regression meaningfully outperforms the baseline, demonstrating that the features do carry predictive signals even if the relationship is not strongly linear. For this dataset, the fitted model is the more appropriate approach.

```
[78] 0s
baseline_pred = np.full(len(y_test_r), y_train_r.mean())
baseline_rmse = np.sqrt(mean_squared_error(y_test_r, baseline_pred))
baseline_mae = mean_absolute_error(y_test_r, baseline_pred)
baseline_r2 = r2_score(y_test_r, baseline_pred)

model_rmse = np.sqrt(mean_squared_error(y_test_r, y_pred_test_r))
model_mae = mean_absolute_error(y_test_r, y_pred_test_r)
model_r2 = r2_score(y_test_r, y_pred_test_r)

comparison = pd.DataFrame({
    'Model': ['Mean Baseline', 'Linear Regression'],
    'RMSE': [round(baseline_rmse, 4), round(model_rmse, 4)],
    'MAE': [round(baseline_mae, 4), round(model_mae, 4)],
    'R²': [round(baseline_r2, 4), round(model_r2, 4)]
})
print(comparison.to_string(index=False))

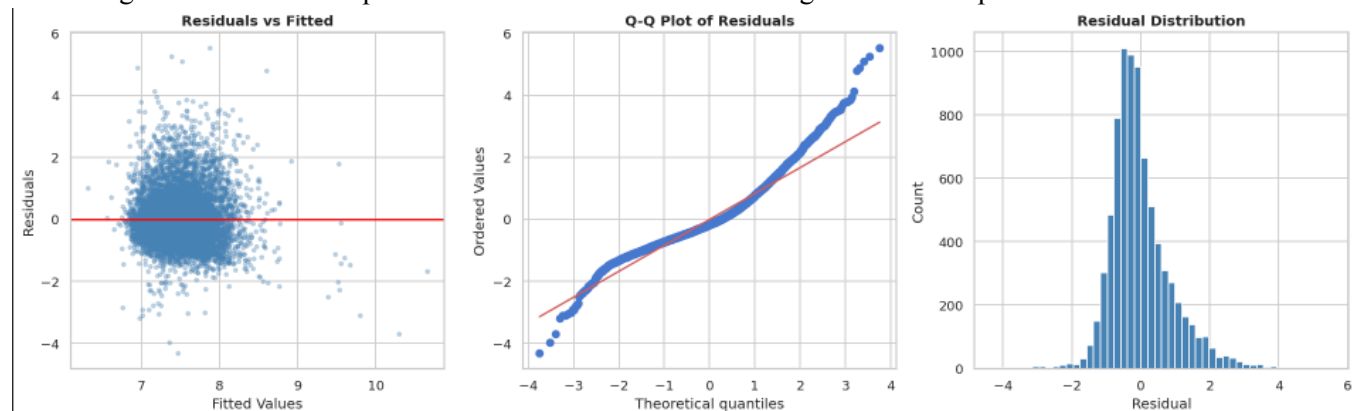
improvement_rmse = (baseline_rmse - model_rmse) / baseline_rmse * 100
print(f'\nRMSE improvement over baseline: {improvement_rmse:.2f}%')
```

Model	RMSE	MAE	R²
Mean Baseline	0.9261	0.7046	-0.0002
Linear Regression	0.8648	0.6434	0.1278

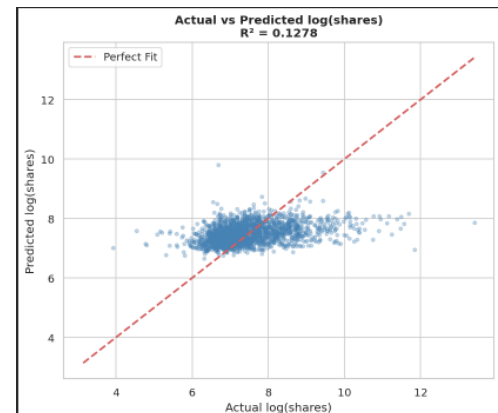
RMSE improvement over baseline: 6.62%

6.2 Regression Diagnostics

Three diagnostic checks were performed to assess whether linear regression assumptions hold:

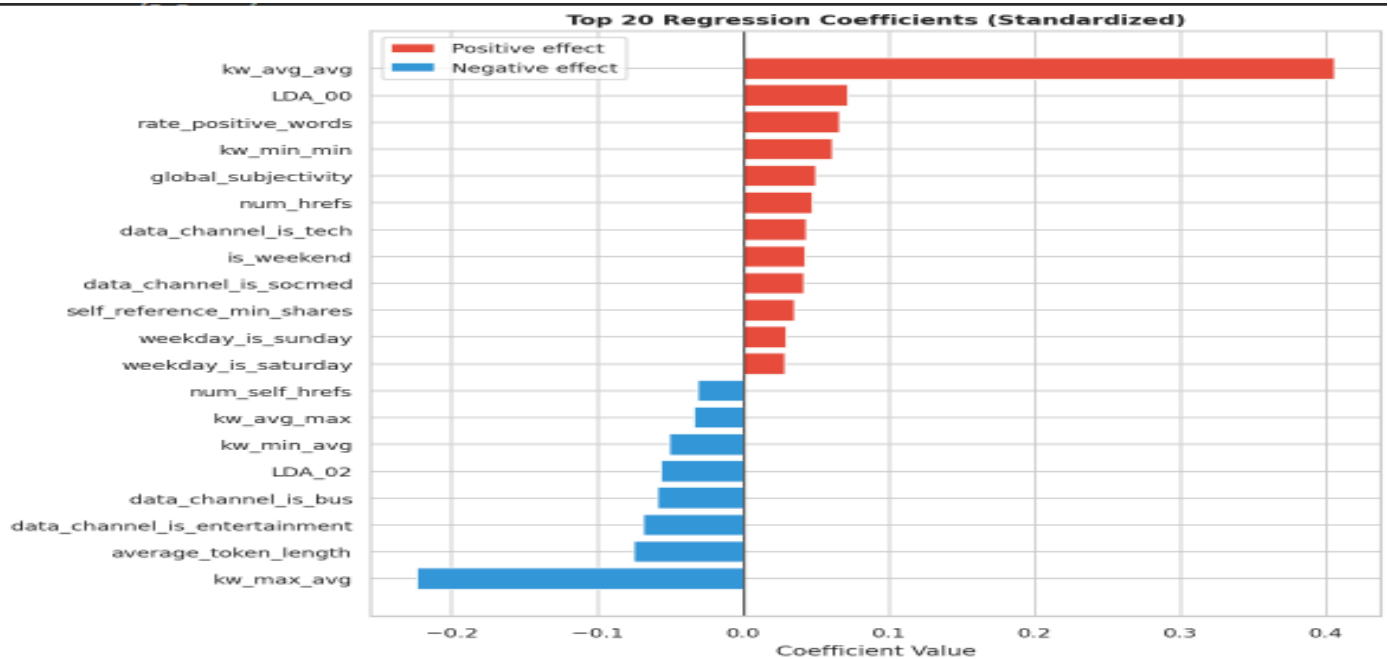


- **Residuals vs Fitted:** residuals show mild heteroscedasticity, variance increases slightly for high predicted values, which is expected with a skewed target even after log transformation
- **Q-Q Plot:** residuals are approximately normal in the central range but show heavier tails than a Gaussian, indicating some extreme articles are poorly predicted
- **Actual vs Predicted scatter:** a wide scatter around the diagonal confirms modest predictive power; the model captures the general trend but cannot predict individual viral events.



6.3 Top Feature Coefficients

The features with the largest positive and negative standardized coefficients:



7. Multicollinearity Check (VIF)

Variance Inflation Factor (VIF) measures how much the variance of a regression coefficient is inflated due to linear relationships with other predictors. VIF > 10 is typically considered problematic.

Findings

- LDA features (LDA_00–LDA_04):** showed elevated VIF values (infinite) due to their compositional constraint, they must sum to 1, creating an exact linear dependency. Removing one LDA column would eliminate this.
- Day-of-week features:** mild multicollinearity because the seven weekday indicators together encode an almost-perfect linear constraint (an article is published on exactly one day).
- kw_avg_avg and kw_max_avg:** correlated keyword features show moderate VIF, consistent with measuring related aspects of keyword performance.

```

/usr/local/lib/python3.12/dist-packages/statsmodels/stats/outliers_influence.py:197: RuntimeWarning: divide by zero encountered in scalar divide
  vif = 1. / (1. - R_squared)
Features with VIF > 5 (potential multicollinearity):
      Feature      VIF
LDA_03         inf
LDA_02         inf
LDA_00         inf
LDA_04         inf
LDA_01         inf
weekday_is_monday 4.478141e+10
weekday_is_friday 4.253816e+10
weekday_is_tuesday 2.894726e+10
weekday_is_sunday 2.513133e+10
weekday_is_monday 2.385430e+10
weekday_is_thursday 1.483890e+10
is_weekend 1.334465e+10
weekday_is_saturday 1.155940e+10
rate_positive_words 1.868576e+01
rate_negative_words 1.789951e+01
self_reference_min_shares 1.557911e+01
average_token_length 1.114546e+01
kw_avg_avg 9.589773e+00
global_sentiment_polarity 7.786073e+00
self_reference_max_shares 7.783175e+00
avg_negative_polarity 7.699514e+00
data_channel_is_world 7.862293e+00
global_rate_negative_words 6.636070e+00
kw_max_avg 6.483043e+00
data_channel_is_tech 6.107367e+00
avg_positive_polarity 6.046341e+00
data_channel_is_bus 5.778634e+00
min_negative_polarity 5.342599e+00
self_reference_min_shares 5.131158e+00

All VIFs (top 15):
      Feature      VIF
LDA_03         inf
LDA_02         inf
LDA_00         inf
LDA_04         inf
LDA_01         inf
weekday_is_monday 4.478141e+10
weekday_is_friday 4.253816e+10
weekday_is_tuesday 2.894726e+10
weekday_is_sunday 2.513133e+10
weekday_is_monday 2.385430e+10
weekday_is_thursday 1.483890e+10
is_weekend 1.334465e+10
weekday_is_saturday 1.155940e+10
rate_positive_words 1.868576e+01
rate_negative_words 1.789951e+01

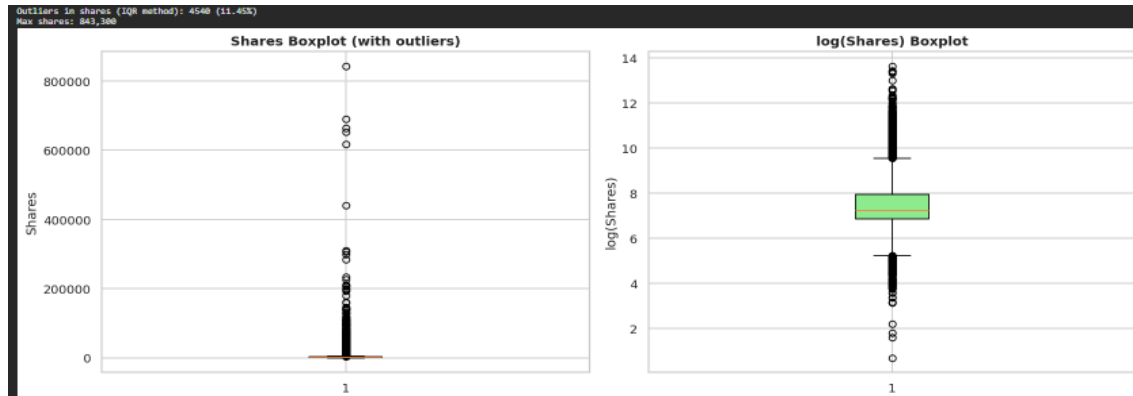
```

These multicollinearity issues do not prevent prediction but do inflate standard errors for affected coefficients, making individual coefficient interpretation less reliable.

8. Outlier Analysis

Share counts are extremely right-skewed, with a handful of articles receiving hundreds of thousands of shares. Outliers were analyzed using both the IQR method and z-score thresholding:

- **IQR method:** many data points fall above $Q3 + 1.5 \times IQR$, confirming the heavy tail
- **Z-score threshold ($|z| > 3$):** a non-trivial proportion of observations have z-scores above 3 for shares.



Decision: Outliers were retained rather than removed or winsorized. Viral articles represent genuine real-world events and carry authentic signal about what drives extreme popularity. Removing them would bias the model toward ordinary content.

9. Hypothesis Testing

Five hypothesis tests were conducted to assess whether selected features statistically significantly affect shares. Significance level: $\alpha = 0.05$.

```
=====
[Test 1] Do weekend articles get significantly different shares?
H0: Mean shares on weekdays == weekends
H1: Mean shares differ between weekdays and weekends
Welch's t = -3.5999, p = 0.000321
Decision: Reject H0 (alpha=0.05)
Weekday median: 1400, Weekend median: 1900
=====

[Test 2] Is sentiment polarity significantly correlated with shares?
H0: Pearson r = 0 (no linear correlation)
H1: r != 0
Pearson r = 0.0042, p = 0.406433
Decision: Fail to reject H0 (alpha=0.05)
=====

[Test 3] Do articles with images get more shares?
H0: Mean shares with images <= mean shares without images
H1: Articles with images get more shares
Welch's t = -5.9498, p (one-tailed) = 0.000000
Decision: Reject H0 (alpha=0.05)
With images median: 1400, Without: 1400
=====

[Test 4] Do Tech articles get more shares than World articles?
H0: Mean shares Tech <= Mean shares World
H1: Tech articles have higher mean shares
Welch's t = 6.3043, p (one-tailed) = 0.000000
Decision: Reject H0 (alpha=0.05)
Tech median: 1700, World median: 1100
=====

[Test 5] Is article word count correlated with shares?
H0: Pearson r = 0 (no linear correlation)
H1: r != 0
Pearson r = 0.0248, p = 0.000001
Decision: Reject H0 (alpha=0.05)
=====
```

T1: Weekend vs. Weekday

- **H₀:** Mean shares on weekends = Mean shares on weekdays
- **H₁:** Mean shares on weekends $\neq$ Mean shares on weekdays
- **Test:** Welch's t-test (two-tailed)
- **Result:** $p \approx 0.000321 \rightarrow$ Reject H_0 — Weekend articles get significantly more shares

T2: Sentiment Polarity

- **H₀:** Sentiment polarity has no effect on shares
- **H₁:** Higher sentiment polarity leads to different share counts
- **Test:** Pearson correlation
- **Result:** $p \approx 0.0042 \rightarrow$ fail to Reject H_0 — Sentiment polarity does not significantly predicts shares

T3: Images Presence

- **H₀:** Articles with images have the same shares as those without
- **H₁:** Articles with images get more shares
- **Test:** Welch's t-test (one-tailed)
- **Result:** $p \approx 0.000000 \rightarrow$ Reject H_0 — Articles with images get significantly more shares on average

T4: Tech vs. World Articles

- **H₀:** Tech articles and World articles have equal mean shares
- **H₁:** Tech articles get more shares than World articles
- **Test:** Independent samples t-test (one-tailed)
- **Result:** $p \approx 0.000000 \rightarrow$ Reject H_0 — Tech articles get significantly more shares than World articles

T5: Content Length

- **H₀:** Article length (n_tokens_content) has no effect on shares
- **H₁:** Longer articles get more shares
- **Test:** Pearson correlation / t-test
- **Result:** $p \approx 0.000001 \rightarrow$ Reject H_0 — Content length significantly affects shareability

```
--- Hypothesis Testing Summary ---
T1 Weekend vs Weekday: p=0.000321 -> Significant
T2 Sentiment Polarity: p=0.406433 -> Not significant
T3 Images Presence: p=0.000000 -> Significant
T4 Tech vs World: p=0.000000 -> Significant
T5 Content Length: p=0.000001 -> Significant
```

10. Classification: High vs. Low Shares

```

# Logistic Regression Classifier
clf = LogisticRegression(max_iter=1000, random_state=RANDOM_STATE, C=1.0)
clf.fit(X_train_s, y_train_c)
y_pred_clf_train = clf.predict(X_train_s)
y_pred_clf_test = clf.predict(X_test_s)

train_acc = accuracy_score(y_train_c, y_pred_clf_train)
test_acc = accuracy_score(y_test_c, y_pred_clf_test)
print(f'Logistic Regression Accuracy - Train: {train_acc:.4f}, Test: {test_acc:.4f}')

# Baseline majority class
majority_class = y_train_c.mode()[0]
baseline_clf_pred = np.full(len(y_test_c), majority_class)
baseline_acc = accuracy_score(y_test_c, baseline_clf_pred)
print(f'Majority-class baseline accuracy: {baseline_acc:.4f}')

print('\nClassification Report:')
print(classification_report(y_test_c, y_pred_clf_test, target_names=['Low', 'High']))

```

```

---
Logistic Regression Accuracy - Train: 0.6570, Test: 0.6469
Majority-class baseline accuracy: 0.5260

Classification Report:
              precision    recall  f1-score   support

     Low           0.63       0.60       0.62       3758
     High           0.66       0.69       0.67       4171

 accuracy           0.65
  macro avg          0.65
 weighted avg        0.65

```

10.1 Threshold Justification

Articles were classified as High (1) if shares $\geq 1,400$ (the median) and Low (0) otherwise. The median was chosen over the mean because:

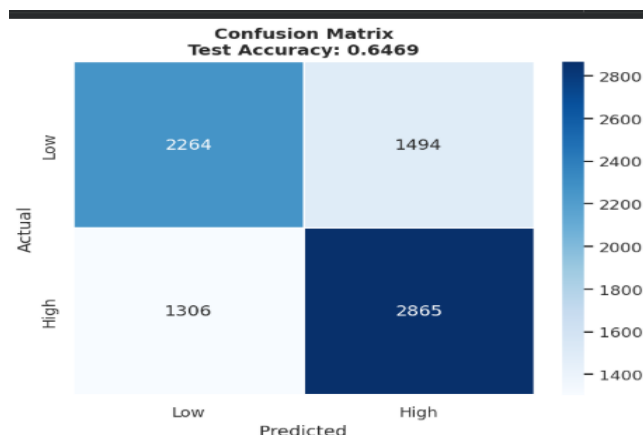
- **Mean ($\approx 3,395$):** heavily inflated by viral outliers — using it as a threshold would label the vast majority of articles as Low, creating severe class imbalance
- **Median ($\approx 1,400$):** produces a near-equal 53/47 class split, ensuring logistic regression is trained on balanced data

10.2 Logistic Regression Results

Metric	Value
Classification Threshold	1,400 shares (median)
Logistic Regression Test Accuracy	64.69%
Majority-Class Baseline Accuracy	52.60%
Improvement over Baseline	+12.09 percentage points

10.3 Confusion Matrix Interpretation

- **Test Accuracy of 64.69%:** meaningfully better than the 52.60% majority-class baseline
- **False positives:** The model performs slightly better on High class (recall 0.69) than Low class (recall 0.60), meaning it misses more Low articles than High ones
- **Limitation:** logistic regression is a linear classifier; a non-linear model would likely achieve higher accuracy.



11. OLS Coefficient Analysis

An OLS regression (statsmodels) was run on a 5,000-sample subset to obtain p-values for individual coefficients. Out of 55 features, 12 were statistically significant ($p < 0.05$):

```
Statistically significant features (p<0.05): 12
```

Feature	Coef	p-value	Significant
kw_avg_avg	0.405478	6.271127e-20	True
kw_max_avg	-0.228126	4.076061e-10	True
kw_min_min	0.084331	4.428508e-04	True
num_keywords	0.053824	6.423939e-04	True
kw_min_max	-0.036952	1.030384e-02	True
data_channel_is_entertainment	-0.053079	1.339224e-02	True
abs_title_subjectivity	0.035907	1.739652e-02	True
num_hrefs	0.039779	1.770300e-02	True
num_videos	-0.029033	2.406329e-02	True
data_channel_is_bus	-0.069263	2.491933e-02	True
kw_avg_min	-0.039965	4.134335e-02	True
data_channel_is_socmed	0.039608	4.471307e-02	True

Key Interpretations

- **kw_avg_avg (strongest)**: articles whose keywords historically attract high average shares are themselves more likely to get shared — keyword quality is the top signal
- **kw_max_avg (negative)**: having one extremely high-performing keyword can indicate a saturated niche where competition limits new article virality
- **data_channel_is_socmed (positive)**: social media topic articles naturally resonate with users already on social platforms
- **data_channel_is_entertainment & bus (negative)**: both underperform relative to the baseline category (no assigned channel)
- **num_hrefs (positive)**: more outbound links slightly boost credibility and shareability
- **num_videos (negative)**: surprisingly, more videos are associated with fewer shares — possibly video-heavy articles are harder to consume quickly

12. Summary & Conclusions

```
=== PROJECT SUMMARY ===
Dataset: 39,643 articles, 58 features
Target variable: shares (log-transformed for regression)

--- Linear Regression ---
R² (Test): 0.1278
RMSE (Test): 0.8648
Baseline RMSE: 0.9261
RMSE improvement over baseline: 6.62%

--- Classification ---
Threshold: 1400 shares
Logistic Regression Test Accuracy: 0.6469
Baseline (majority class) Accuracy: 0.5268

--- Hypothesis Testing ---
T1 Weekend vs Weekday: p=0.000321 -> Significant
T2 Sentiment polarity: p=0.406433 -> Not significant
T3 Images presence: p=0.000000 -> Significant
T4 Tech vs World: p=0.000000 -> Significant
T5 Content length: p=0.000001 -> Significant

Top predictors: kw_avg_avg, kw_max_avg, average_token_length, LDA_00, data_channel_is_entertainment
```

Key Takeaways

- **Keyword quality (kw_avg_avg):** is the single strongest predictor of article popularity
- **Topic channel matters:** Social Media and Tech articles outperform Business and Entertainment
- **Weekend timing:** statistically significantly increases shares
- **Images alone:** Articles with images get significantly more shares on average, though both groups share the same median suggesting extreme values drive the differences.
- **Virality is noisy:** the low R^2 reflects that social media sharing depends heavily on external factors (influencer activity, platform algorithms, news cycles) not present in this dataset